

SHABAB AHMED

PH.D. CANDIDATE IN ECONOMICS
UNIVERSITY OF WASHINGTON

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Ph.D. Candidate in Economics with expertise in structural modeling, contract theory, and applied econometrics. Experienced in developing dynamic models, analyzing large-scale survey data, and translating research into insights that inform business and policy decisions.

EXPERIENCE

Economist Intern, Amazon

JUL 2025 – OCT 2025

BELLEVUE, WA

- Built structural demand models on large-scale survey data to estimate customer preferences for Project Kuiper, with findings that helped inform the team's pricing strategy.
- Used a tree-based feature-importance model to identify the key drivers of customer response to determine which inputs entered the structural model.
- Built a simulation platform to evaluate counterfactual business scenarios across multiple pricing and product configurations.
- Authored and presented a technical and business report that translated econometric findings into recommendations for senior stakeholders.

Graduate Teaching Assistant & Independent Instructor

2019 – PRESENT

UNIVERSITY OF WASHINGTON

- Independent Instructor for Game Theory, with full responsibility for course delivery, assignments, and assessment.
- Teaching Assistant for Graduate Microeconomic Theory and undergraduate courses, leading discussion sections and review sessions.

EDUCATION

Ph.D. in Economics, University of Washington

EXPECTED NOV 2026

DISSERTATION

Dynamic moral hazard and incentive design — how institutions should adapt when preferences evolve over time, applied to NFL draft policy

High-dimensional demand estimation — using Lasso and Elastic Net regularization with debiasing corrections to estimate elasticity under collinearity, validated via Monte Carlo simulation

Dynamic moral hazard under a fixed budget — how agents allocate spending across time under a budget constraint, and how this can explain patterns like budget hoarding and end-of-year spending spikes

Passed general examination; presented research in departmental seminars.

M.A. in Economics, University of Washington DEC 2022 · GPA 3.89

M.S. in Applied Mathematics, University of Washington JUN 2020 · GPA 3.87

B.A. in Mathematics and Economics, Colby College MAY 2019 · GPA 3.93

Magna Cum Laude, Phi Beta Kappa, Distinction in Economics & Mathematics, Dean's List (6 semesters).

SELECTED PROJECTS

NBA Decade Comparison: PCA and Machine Learning Analysis WINTER 2020

- Built classifiers on 13,000 player-season observations (~95 features scraped from Basketball Reference). Used PCA and LDA to predict player position and decade, and L1-regularized logistic regression to predict Win Shares quartile.
- Achieved 72.6% accuracy on the Win Shares model, validated via 50-fold cross-validation. Raw features outperformed PCA-compressed inputs by over 25 points, which shaped the final feature strategy.

Game Theory in Cricket & Attacks on RSA 2019

- Built a formal game-theoretic model of bowler-batsman strategy, validated against an original dataset built from match commentary.
- Implemented and analyzed multiple RSA cryptographic attacks in Python.

TECHNICAL SKILLS

METHODS	Structural demand estimation, dynamic optimization, Lasso/Elastic Net regularization, debiased ML, causal inference (IV, DiD, panel data), PCA, LDA, logistic regression, tree-based models
TOOLS	Python, R, Stata, SQL, MATLAB, Sage, Mathematica, LaTeX, Qualtrics, Excel
LANGUAGES	English (fluent), Bengali (fluent), Hindi (conversational)